

This assignment will not be graded and is only for practice.

Level 1

1) The dimension of a vector space is the :

1 point

- ☐ Cardinality of any spanning set.
- ☐ Cardinality of a basis.
- ☐ Cardinality of minimal spanning set.
- ☐ Cardinality of maximal linearly independent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Cardinality of a basis.

Cardinality of minimal spanning set.

Cardinality of maximal linearly independent set.

2) Match the sets of vectors in column A with their properties of linear dependence or

1 point

independence in column B and the dimension of the vector spaces in column C spanned by the sets.

	Set of vectors (Column A)		Linear dependence or independence (Column B)		Dimension of the vector space spanned by the set (Column C)
a)	$\{(1, 0, 0), (0, 1, 0), (1, 1, 1), (1, 0, 1)\}$	i)	Linearly independent	1)	1
b)	$\{(1, 0, -1), (0, -1, 0), (-1, 1, -1)\}$	ii)	Linearly dependent	2)	2
c)	$\{(2, 3, 6), (\frac{1}{3}, \frac{1}{2}, 1), (4, 6, 12)\}$	iii)	Linearly dependent	3)	3
d)	$\{(1, 0, -\frac{1}{6}), (0, 1, -\frac{1}{3}), (3, 2, -\frac{7}{6})\}$	iv)	Linearly dependent	4)	3

Table : M2W4AQ3

- ☐ $a \rightarrow ii \rightarrow 3, b \rightarrow iii \rightarrow 2$
- ☐ $c \rightarrow iii \rightarrow 2, d \rightarrow iv \rightarrow 1$
- ☐ $a \rightarrow ii \rightarrow 4, b \rightarrow i \rightarrow 3$
- ☐ $c \rightarrow iv \rightarrow 1, d \rightarrow iii \rightarrow 2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$a \rightarrow ii \rightarrow 4, b \rightarrow i \rightarrow 3$
 $c \rightarrow iv \rightarrow 1, d \rightarrow iii \rightarrow 2$

3) Choose the set of correct options.

1 point

- ☐ The dimension of the vector space $M_{1 \times 2}(\mathbb{R})$ is 2.
- ☐ The dimension of the vector space $M_{2 \times 1}(\mathbb{R})$ is 1.
- ☐ The dimension of the vector space $M_{3 \times 3}(\mathbb{R})$ is 3.
- ☐ A basis of $M_{2 \times 2}(\mathbb{R})$ is the set $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

The dimension of the vector space $M_{1 \times 2}(\mathbb{R})$ is 2.

A basis of $M_{2 \times 2}(\mathbb{R})$ is the set $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

4) Consider the following sets:

1 point

$V_1 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a symmetric matrix, i.e., } A = A^T\}$
 $V_2 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a scalar matrix}\}$
 $V_3 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a diagonal matrix}\}$

$V_4 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is an upper triangular matrix}\}$

$V_5 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a lower triangular matrix}\}$

All $V_i, i = 1, 2, 3, 4, 5$ are subspaces of the vector space $M_{2 \times 2}(\mathbb{R})$.

Choose the set of correct options.

- ☐ The dimension of V_1 is 3.
- ☐ The dimension of V_2 is 3.
- ☐ The dimension of V_3 is 1.
- ☐ The dimension of V_4 is 3.
- ☐ The dimension of V_5 is 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The dimension of V_1 is 3.

The dimension of V_4 is 3.

The dimension of V_5 is 3.

5) Consider the following two matrices M_1 and M_2 .

1 point

$$M_1 = \begin{bmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Choose the correct set of options.

- ☐ The row rank of both the matrices, M_1 and M_2 , is 2.
- ☐ The row rank of both the matrices, M_1 and M_2 , is 3.
- ☐ The column rank of M_1 is 4, but the column rank of M_2 is 3.
- ☐ The column rank of M_2 is 4, but the column rank of M_1 is 3.
- ☐ The column rank of both the matrices, M_1 and M_2 , is 4.
- ☐ The column rank of both the matrices, M_1 and M_2 , is 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The row rank of both the matrices, M_1 and M_2 , is 3.

The column rank of both the matrices, M_1 and M_2 , is 3.

Level 2

6) Find the dimension of the vector space $V = \{A \mid \text{sum of entries in each row is 0, and } A \in M_{3 \times 2}(\mathbb{R})\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

7) Find the dimension of the vector space $V = \{(x, y, z, w) \mid x + y = z + w, x + w = y + z, \text{ and } x, y, z, w \in \mathbb{R}\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

8) Choose the set of correct options.

1 point

- ☐ If $S \subset \mathbb{R}^2$ contains three vectors, then vectors in S must be linearly independent.
- ☐ If $S \subset \mathbb{R}^3$ contains three vectors, then vectors in S may or may not be linearly independent.

- ☐ If $S \subset \mathbb{R}^4$ contains five vectors, then vectors in S must be linearly dependent.
- ☐ If $S \subset \mathbb{R}$ contains one vector, then the vector in S must be linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $S \subset \mathbb{R}^3$ contains three vectors, then vectors in S may or may not be linearly independent.

If $S \subset \mathbb{R}^4$ contains five vectors, then vectors in S must be linearly dependent.

9) Find the dimension of the vector space $V = \{A \in M_{3 \times 3}(\mathbb{R}) : A = A^T\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

10) Find the dimension of the vector space $V = \{A \in M_{3 \times 3}(\mathbb{R}) : A = -A^T\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Your score is: 0/10